

Retail Growth & Customer Retention Analytics Platform

Python, SQL, Pandas, Scikit-learn, Market Basket Analysis, Power BI

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Case Study

Business Problem

This project analyzes over 500,000 e-commerce transactions spanning 01/12/2010 to 09/12/2011 from a UK-based online retailer to identify customer value, retention patterns, churn risk, product revenue drivers, and cross-sell opportunities. The central business question was: *Which customers are most valuable, which customers are at risk of churn, what products drive revenue, and what actions should the business take?*

The project was designed as an end-to-end analytics workflow that turns raw transaction-level data into business recommendations for growth, retention, merchandising, and customer lifecycle strategy.

Data and Methodology

The raw dataset included invoice number, product code, product description, quantity, invoice date, unit price, customer ID, and country. The data was cleaned by removing invalid transactions, separating canceled orders, standardizing fields, and creating revenue variables. Two cleaned datasets were created: one full transaction dataset for product, revenue, and country analysis, and one customer-level transaction dataset for segmentation, retention, and churn analysis.

The analysis included five major components:

- **Revenue and product analysis:** Monthly revenue, product revenue, order volume, country revenue, and average order value were calculated.
- **RFM segmentation:** Customers were scored using recency, frequency, and monetary value, then classified into business-friendly segments such as Champions, Loyal Customers, High-Value At Risk, Needs Attention, and Lost or Dormant.
- **Cohort retention analysis:** Customers were grouped by first purchase month to measure how many returned in later months.
- **Churn-risk scoring:** A random forest model was trained using customer behavior features to estimate churn risk and prioritize retention actions.
- **Market basket analysis:** Association rules were used to identify products frequently purchased together using support, confidence, and lift.

The churn-risk model achieved an accuracy of 0.662, AUC of 0.741, precision of 0.580, and recall of 0.768. The model prioritized recall, which is appropriate for retention campaigns where missing an at-risk customer may be more costly than contacting some customers unnecessarily.

Key Findings

Revenue increased strongly into Q4

Monthly revenue generally increased in the later months of the dataset, with November 2011 representing the strongest revenue month. December 2011 was treated as a partial month and was not interpreted as a true month-over-month decline. This pattern suggests that seasonal planning is important for inventory, staffing, and promotional timing.

Champion customers drive the majority of revenue

The RFM analysis showed that the Champion customer segment generated nearly 65% of total revenue. This indicates that a relatively small group of high-value, high-frequency, recent customers is responsible for a major portion of business performance. Protecting this segment should be a priority because small changes in Champion retention could have an outsized revenue impact.

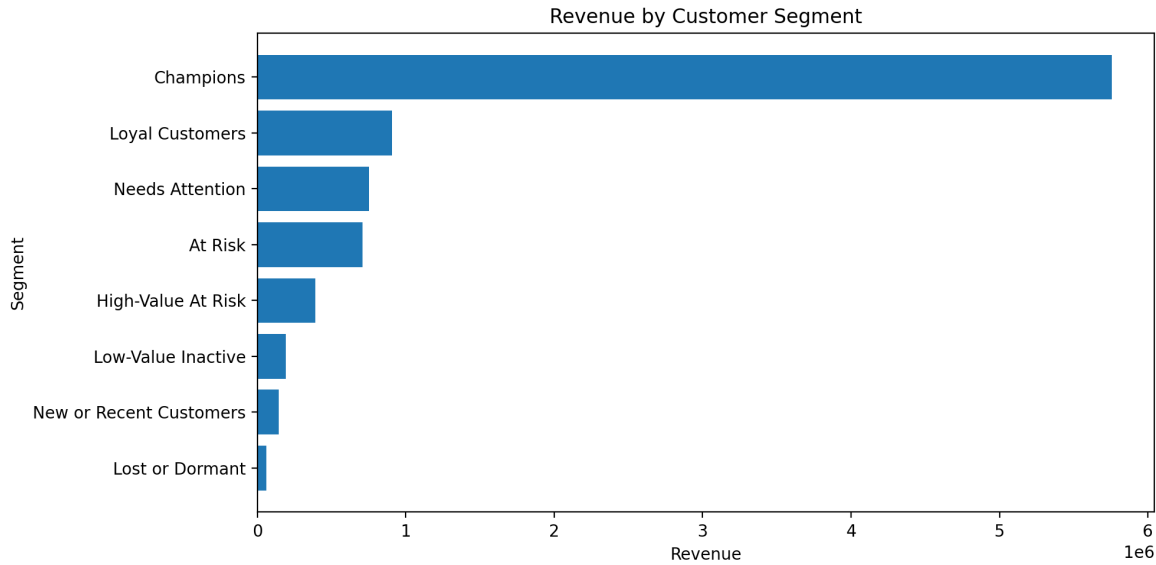


Figure 1: Revenue by customer segment. Champion customers generate the largest share of total revenue, showing that revenue is highly concentrated among the strongest customer relationships.

Retention drops quickly after the first purchase

The cohort retention analysis showed that customer retention declines sharply after the first purchase month. This suggests that the business has a meaningful opportunity to improve second-purchase conversion. A targeted post-purchase journey may be more valuable than broad, untargeted marketing efforts.

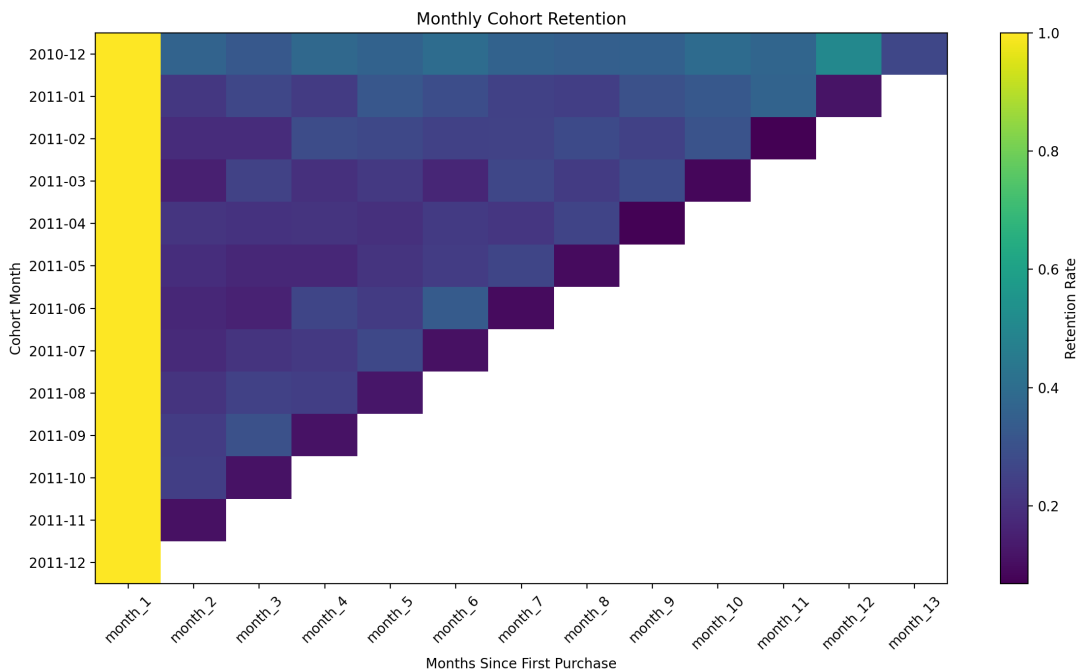


Figure 2: Monthly cohort retention by first purchase month. Most cohorts show a steep decline after the first purchase period, highlighting the importance of second-purchase campaigns and lifecycle marketing.

High-value at-risk customers represent a focused reactivation opportunity

The churn-risk scoring framework identified customers with high historical value but elevated risk of inactivity. These customers are especially important because they have already demonstrated willingness to spend, making them stronger candidates for reactivation than low-value inactive customers.

Repeatable product revenue is concentrated in a small set of products

After filtering out non-product charges, manual adjustments, and one-off outliers, product revenue was concentrated among a small set of repeatable products. The strongest product categories were decorative and home goods, which should receive attention in merchandising, inventory planning, and promotional strategy.

Recommendations

- 1. Prioritize high-value at-risk customers.** Customers with strong historical value and elevated churn risk should receive priority reactivation campaigns. Recommended actions include personalized discounts, targeted emails, loyalty offers, or early access to new products.
- 2. Improve second-purchase conversion.** Since retention drops sharply after the first purchase month, the business should create a post-purchase lifecycle campaign designed to encourage a second purchase within 30 to 60 days. This could include product recommendations, limited-time offers, or bundle suggestions.
- 3. Protect Champion customers.** Champion customers should receive loyalty-focused engagement, such as exclusive offers, personalized recommendations, or early product access. Because this segment drives the largest revenue share, retaining these customers should be treated as a core business priority.
- 4. Use product bundles to increase average order value.** Market basket analysis identified products that are frequently purchased together. These products can be tested as bundles, cross-sell recommendations, or co-promoted items on product pages and email campaigns.
- 5. Improve seasonal planning.** The revenue trend suggests strong Q4 demand. The business should prepare inventory, promotions, and operational capacity ahead of seasonal peaks to reduce stockouts and maximize revenue capture.

Limitations and Next Steps

This analysis was limited by the dataset's historical time range and the absence of customer demographic or marketing campaign data. Churn was defined behaviorally based on future purchase activity within the dataset, rather than through a formal subscription cancellation event. December 2011 was also treated carefully because the dataset only contains a partial month.

Future improvements could include customer lifetime value prediction, demand forecasting, more advanced churn models, A/B testing recommendations for retention campaigns, and a live dashboard connected directly to a database.

Conclusion

This project demonstrates how structured transaction data can be transformed into actionable business strategy. By combining revenue analysis, RFM segmentation, cohort retention, churn-risk scoring, market basket analysis, SQL exports, and dashboard reporting, the project identifies clear opportunities to improve retention, prioritize high-value customers, increase average order value, and support merchandising decisions.